Adobe digital assistant

Adobe Research is reportedly working on a voice-activated digital assistant that can edit pictures for you. **http://dgit.in/AdobeVo**

Basslet wearable

Basslet is a small, flat black wearable device that you wear on your wrist like a watch that acts as a subwoofer. **http://dgit.in/BassletW**

Digital stethoscopes

The stethoscope has been hanging around the neck of doctors for so long that the device has become synonymous with the doctor's profession. One of the most basic tools for diagnosis, the device is also getting advanced, thanks to digital technology. And this is definitely bettering the quality of diagnoses that are being made.

One of the problems with traditional acoustic stethoscopes is that the sound level you get is very low. Using modern technology, digital stethoscopes electronically amplify the body sounds and

An age old device gets a makeover!

overcome the issue. This would require that the acoustic sounds be converted into electronic signals which would be then transmitted through a uniquely designed circuit. These signals would then be processed for the purpose of optimal listening. The circuit would include components which allow the amplification of the energy and also supports listening at different frequencies. Not just that, the sound energy could be digitized, encoded and decoded. Also, ambient noise could be reduced or even completely eliminated and the result can be channelled through headphones or speakers.

Whereas traditional stethoscopes are all based on the same physics, the transducers used in the electronic stethoscopes differ. Probably the least effective (and also the simplest) way to detect sound could be placing a microphone in the chest piece. The drawback here is ambient noise interference. Another method is to place a piezoelectric crystal at the head of a metal shaft-the shaft's bottom would make contact with a diaphragm. The diaphragm would respond to sound waves in the same manner that traditional acoustic stethoscopes do. The difference is that the changes in air pressure would be replaced by changes in an electric field. This way, not only is the sound of an acoustic stethoscope is preserved, but the benefits of amplification can also be had. There would definitely be an electronic tinge to the resultant sound but it would be less ambiguous, thereby helping doctors.

Also, the fact that sounds are in the electronic form means that features like wireless transmission and recording of sound clips are possible. Most digital stethoscopes have an audio output signal which, using a stereo or mono cable connection allows the audio to be transmitted in real time to a supporting app. The supporting software interprets the output for diagnosis.

Also, the audio outputs that are processed using different supporting software packages could be saved as files to be transmitted through email or other modes of communication methods. Certain digital stethoscopes even have audio data output options that could be hooked up to the audio input of a video-conferencing unit. The video-conferencing units would then be the decoders and encoders of the data to be transmitted through a network.

Some digital stethoscopes even allow data to be transmitted through a Bluetooth or wireless interface. Most wireless digital stethoscopes may require Bluetooth receivers.

Some models allow on-board recording and playback on the physical part of the stethoscope itself. Visual output of such parameters like the heart rate and EKG waves are also possible with certain supporting apps.

The buck doesn't stop there for digital stethoscopes. The fact that they convert sound to digital signals means that they can transmit the data either live or in a store-and-forward manner. These devices detect sound using the stethoscope sensor and then convert the sound energy into electricity which would be channelled through a circuit that can amplify it and filter it by frequency before converting the analog data into its digital equivalent. A corresponding digital stethoscope should be on the other end so that the digital signals could be converted into analog signal so that the sound can be processed by the human ear.

A Tricorder – a la Star Trek

Don't get too excited for the idea of diagnosing someone using a tricorder as you see in Star Trek must remain in the realm of science fiction for a while. But certain new researches are pointing to a future in which this could be a reality soon enough. For instance, the idea of using breath tests to diagnose bodily disorders, if made possible would be a relatively inexpensive, not to mention painless and quick method of diagnosis.

Related to this, a team of global researchers recently unveiled a nano array which could identify the chemical signature of 17 various diseases. However, for anything like a tricorder

Science-fiction, soon to be non-fiction!

to be made possible with breath tests alone, the breathalyzers should be able to identify more than a disease at a time. The technologies that have been developed so far are rather limited in scope in this regard. However, the recent research has identified different breath odour signatures of different diseases. Simply put, the composition of the breath that we exhale would vary depending on our state of health. Now, if there exists a breathalyzer (a la tricorder) that could analyse these compositions correctly, we could have ourselves a 'cool' medical device.

New frontiers seem to be just a breath away.